Corporate Bankruptcy Prediction

UE24CS352A - Machine Learning | Concise submission summary

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Problem and dataset

Predict bankruptcy within three years from 64 financial ratios. The UCI Polish Companies Bankruptcy third-year file contains 10,503 statements and 495 positive labels. Removing 87 exact duplicates leaves 10,416 statements and 493 positives (4.73%). Missing feature values are retained for model-specific handling.

Method and implementation

A stratified 80/20 split (seed 42) reserves 2,084 test statements, including 99 positives. Logistic regression, Gaussian naive Bayes, RBF SVM, random forest, extra trees, and histogram gradient boosting are compared using identical three-fold development splits. Learned preprocessing stays inside each training fold. Mean cross-validation average precision selects the model; development out-of-fold F2 selects the decision threshold.

Histogram gradient boosting leads with mean development AP 0.619 (fold SD 0.037). The selected configuration uses 250 iterations, learning rate 0.07, 15 leaves, L2 regularization 1, balanced class weights and native missing-value handling. The frozen threshold is 0.1980.

Verified test results

Measure	Selected model	Always non-bankrupt
Recall	84.8% (84 / 99)	0.0%
Precision	34.3%	0.0%
Average precision / ROC AUC	0.750 / 0.968	0.048 / 0.500
Accuracy	91.55%	95.25%
False positives / false negatives	161 / 15	0 / 99

The default 0.5 threshold gives 68.7% recall and 65.4% precision. Lowering it to the development-selected value gains 16 detections and 125 false alarms. Although 0.5 has higher test F2, the selected threshold remains fixed to preserve the evaluation protocol.

Conclusion and limitations

The model provides a useful ranking signal on this historical sample, with substantial false alarms. Scores are uncalibrated. Company IDs are unavailable, so exact duplicate removal cannot ensure company-level independence. A random holdout does not establish performance in later years or other countries. Future work should test group/temporal splits and threshold stability.

Deliverables include reproducible source code, an executed notebook, saved model, split and prediction evidence, evaluation figures, automated tests, a CSV-capable local demo, and presentation materials.

Sources

Tomczak (2016), UCI Polish Companies Bankruptcy, doi:10.24432/C5F600 (CC BY 4.0). Scikit-learn evaluation and cross-validation documentation. Background: supplied Patil report and github.com/codeatri/Bankruptcy_Prediction. All results above come from this independent implementation.